sustainability

MDPI

Review

Generative AI for Consumer Behavior Prediction: Techniques and Applications

Mitra Madanchian

Department of Arts, Communications and Social Sciences, University Canada West, Vancouver, BC V6Z 0E5, Canada; mitra.madanchian@gmail.com

Abstract: Generative AI techniques, such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and transformers, have revolutionized consumer behavior prediction by enabling the synthesis of realistic data and extracting meaningful insights from large, unstructured datasets. However, despite their potential, the effectiveness of these models in practical applications remains inadequately addressed in the existing literature. This study aims to investigate how generative AI models can effectively enhance consumer behavior prediction and their implications for real-world applications in marketing and customer engagement. By systematically reviewing 31 studies focused on these models in e-commerce, energy data modeling, and public health, we identify their contributions to improving personalized marketing, inventory management, and customer retention. Specifically, transformer models excel at processing complicated sequential data for real-time consumer insights, while GANs and VAEs are effective in generating realistic data and predicting customer behaviors such as churn and purchasing intent. Additionally, this review highlights significant challenges, including data privacy concerns, the integration of computing resources, and the limited applicability of these models in real-world scenarios.

Keywords: generative AI; consumer behavior prediction; transformer models; recommendation systems; sentiment analysis

check for updates

Citation: Madanchian, M. Generative AI for Consumer Behavior Prediction: Techniques and Applications. *Sustainability* **2024**, *16*, 9963. https://doi.org/10.3390/su16229963

Received: 21 September 2024
Revised: 4 November 2024
Accepted: 5 November 2024
Published: 15 November 2024

Copyright: © 2024 by the author. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Research on consumer behavior has many facets. A state-of-the-art review of consumer behavior still needs to be improved despite the availability of contemporary reviews on several aspects of consumer behavior [1–3]. Technology, individual awareness, and cultural changes influence consumer behavior, complicating marketing planning [4,5]. Consumers have various opinions about the same goods. Thus, they behave differently [6]. Consumers value corporate social responsibility for sustainable development [7,8]. Internal and external influences shape consumer behavior. Internal elements include economic and psychological situations, whereas external influences include social and cultural factors [9].

Researchers and marketers study market consumer behavior since it is vital to economics. People buy items and services based on requirements, preferences, budget limits, and advertising and promotions [10]. There are various theories and models of consumer behavior. Brand loyalty, friend and family referrals, social influence, and peer pressure can also affect consumers. Depending on consumers and situations, some of these elements may be more or less essential [11]. Marketing strategy and customer behavior interact dynamically and cyclically. Research on consumer behavior can yield insights that help marketers create more specialized campaigns that better meet the requirements and expectations of their target audience. These insights can then impact how consumers behave in the future [12].

Marketers make investments across various media channels to sway customer behavior. Every media platform has a unique composition of advertisements that distinctively

engage viewers. Consumers' media habits have changed due to digitalization [13]. Integrating artificial intelligence (AI) tools into mainstream use has significantly altered how individuals work, learn, and enhance their quality of life over the past few years [14,15]. The emergence of generative AI has profoundly impacted several domains, including science, literature, and the arts. Generally, generative AI describes systems that can learn patterns from existing data to produce new material, such as text, graphics, or music. Models like Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and more recent developments like diffusion models and large language models (LLMs) are essential to this technology [16,17]. Applications ranging from complicated problem-solving in research contexts to automated content generation have benefited greatly from the advancement of these models.

Generative models, like GANs, are made to learn the underlying probability distribution of the data in contrast to their descriptive counterparts.Their main objective is to create new data samples that closely mimic the patterns found in the training data [18,19]. GANs, VAEs, and transformers are generative models that have demonstrated the potential to generate realistic data and predict complex consumer behaviors. This has resulted in more personalized marketing efforts and enhanced customer satisfaction [20,21].

With generative AI, fresh data—such as writing, music, and images—can be created automatically, unlike conventional tasks like regression and classification. The generative model, which simulates the possible data distribution and produces new data comparable to the original data, is an essential part of generative AI [22]. The potential of generative AI tools—especially LLMs like ChatGPT—has completely changed how research is discovered and summarized. Researchers and data scientists can work more efficiently thanks to these tools, which speed up literature searches and make it possible to extract important insights from large datasets. However, because it is so simple to create content, generative AI has also spurred conversations about academic integrity because it raises questions about the validity and uniqueness of scholarly work [23].

There are obstacles to generative AI. Risks include the possibility of "hallucinations", where AI generates false or deceptive information [24,25]. Understanding these technologies' requirements, assessment criteria, and the moral ramifications of using them is becoming increasingly important as they develop [16,17]. The current conversation on generative AI highlights the need for responsible deployment, meaning that even with the technology's amazing potential, it must be applied morally and successfully in various contexts [23]. Despite the optimistic applications of generative AI, there needs to be more studies in the literature regarding the methodologies used to predict consumer behavior and its comprehensive evaluation. The technological aspects of generative AI, such as model architectures and training requirements, are frequently the focus of current research. However, the practical implications of generative AI in consumer insights and market research should be more well considered. The full realization of generative AI's potential in this domain is impeded by challenges such as explainability, robustness, and ethical considerations, which still need to be explored. It is imperative to address these gaps to create frameworks that can effectively incorporate generative AI into consumer behavior analysis, enabling businesses to navigate the intricacies of contemporary consumer dynamics [16,17,26].

This comprehensive review examines GANs, VAEs, and transformers for consumer behavior prediction. Marketing optimization, personalized recommendations, customer segmentation, and sentiment analysis will be reviewed as consumer behavior prediction is important. Empirical research, case studies, and theoretical publications will assess generative AI use's performance, problems, and ethical implications in the real world. This review synthesizes findings from various sources to provide a thorough picture of how advanced AI models are used in consumer behavior prediction, to identify literature gaps, and to suggest future research. The evaluation will also discuss how various AI techniques may affect organizations, including potential implementation hurdles and solutions. This systematic review addresses the following research questions:

- What are the primary themes and trends in the current applications of AI for consumer behavior prediction, and how are these applications transforming consumer insights across industries?
- How do different AI models compare in effectiveness and suitability for consumer behavior prediction and what factors influence their optimal application in various contexts?
- What are the major challenges and limitations associated with integrating AI in consumer behavior prediction, including ethical, technical, and practical considerations, and how can these challenges be mitigated?

2. Theoretical Background

AI technology has significantly transformed consumer behavior analysis, impacting key elements such as personality, engagement, trust, and decision making across various digital touchpoints [27]. Studies using machine learning and statistical methods have advanced understanding in this field [28–30].

The foundational "Theory of Buyer Behavior" by Howard and Sheth (1969) [31] introduced methods for predicting consumer behavior through interdisciplinary analytical techniques, with decision tree algorithms being used to simplify these predictions since 1987 [32]. Psychological theories, including the theory of planned behavior (TPB), offer frameworks for understanding behavioral prediction [33].

Generative AI models, such as Flow-Based Models, Diffusion Models, GANs, and VAEs, each bring unique strengths and limitations to this analysis [34]. GANs, introduced by Goodfellow et al., are extensively studied for their ability to generate synthetic data, supporting tasks from data augmentation to consumer behavior prediction when real data are limited or imbalanced [34]. VAEs, presented by Kingma [35], incorporate probabilistic inference to produce new data samples based on latent consumer behavior patterns, improving predictions through probabilistic encoding and sampling.

2.1. Overview of Generative AI Models

While AI uses machine learning and NLP to evaluate complicated consumer data and produce more nuanced insights, traditional approaches rely on basic demographic analytics [36]. Businesses can better forecast customer preferences and actions thanks to AI tools like sentiment analysis and clustering [15,36,37]. Although AI has the potential to be revolutionary, it also poses issues with algorithmic biases and data privacy. Therefore, innovation and moral behavior must be balanced [38].

Generative models seek to understand the data's underlying distribution, enabling them to generate new instances that resemble the training data, unlike discriminative models that classify data points [39,40]. Generative modeling provides the foundation for this capacity since it allows one to generate synthetic samples depending on learned correlations using joint probability distributions, therefore generating mathematical expressions. Figure 1 illustrates the variety of applications for generative AI.

Generative AI models often utilize complex architectures, particularly deep neural networks (DNNs). Notable examples are shown in Table 1.

Table 1. Some AI model architectures and their applications.

Architecture	Overview	Functionality	Primary Use Cases	Reference(s)
Generative Pre-trained Transformers (GPT)	Models designed for NLP tasks	Capable of generating coherent and contextually relevant text	NLP, Text Generation	[40,41]
Diffusion Models	Models used for visual content creation	Iteratively refine random noise into coherent images	Image Generation, Visual Content Creation	[39]
Multimodal Models	Models that can handle multiple data types (e.g., text and images)	Process and generate content across different data types, enhancing versatility	Cross-Modal Content Generation, Unified AI Tasks	[42,43]

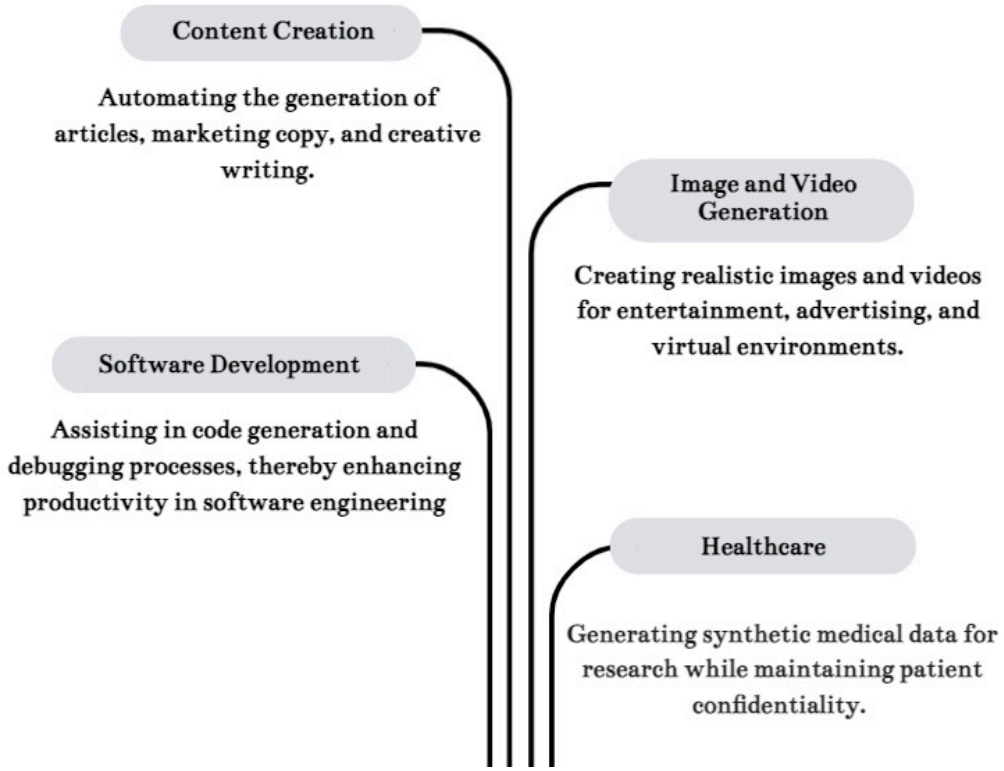


Figure 1. Generative AI applications.

2.2. The Role of AI in Consumer Behavior Prediction

Traditionally, consumer behavior prediction has depended on statistical techniques and rule-based systems, which, although useful to some degree, frequently need help to reflect the complexity and dynamism of contemporary consumer markets. Usually requiring considerable feature engineering, traditional techniques such as logistic regression or decision trees have limited capacity to represent non-linear connections and high-dimensional data. These constraints have spurred research on more advanced AI-driven methods capable of automatically learning intricate patterns from big datasets without much human involvement [44].

AI-driven methods, particularly those grounded on generative models, have greatly improved the accuracy and depth of consumer behavior forecasts. Using big-scale datasets and sophisticated machine learning algorithms, these models can find latent trends and connections that conventional approaches would overlook. GANs can be used, for instance, to create realistic customer profiles based on current data, enabling marketers to replicate how new goods or services could be welcomed by various populations [45]. In the same vein, VAEs can replicate the underlying latent variables influencing consumer preferences, therefore offering a more thorough understanding of why individuals choose what they do and how these decisions may vary in reaction to outside events [46].

The growing availability of consumer data and the creation of AI systems competent in processing and evaluating these data at scale have driven the change in marketing toward data-driven decision making. Generative AI models provide a strong tool for improving conventional consumer behavior models in this environment using more accurate, complex, and pragmatic insights. Transformers have proven particularly useful in recommendation systems; for example, they can accurately predict future behaviors and preferences by analyzing enormous volumes of customer contact data [47–49].

2.3. Theoretical Models of Consumer Behavior

Consumer behavior research is based on many theoretical models that clarify why and how people decide what to buy. Usually including several psychological, social, and

financial elements, these models have driven marketing plans for many years [50–52]. Figure 2 shows the Economic, Psychological, Sociocultural, and Personal Factors models of consumer behavior, each highlighting various factors in consumer decision making.

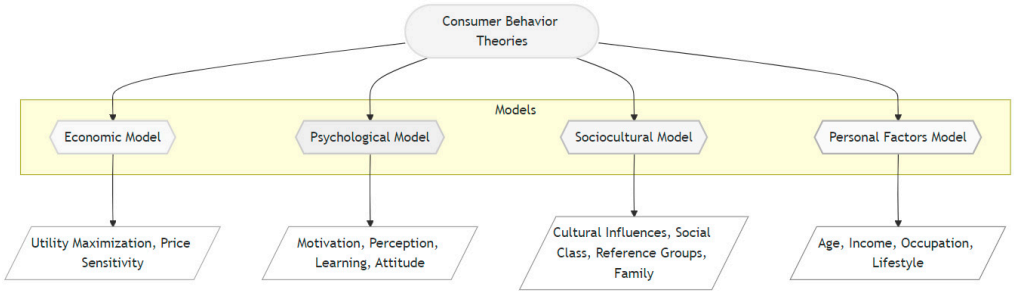


Figure 2. Types of models of consumer behavior.

Presented by Ajzen [53], the TPB is among the most often accepted models. According to the TPB, attitudes toward the activity, subjective norms, and perceived behavioral control shape behavioral intentions—that which drives consumer behavior. By analyzing customer attitudes and the effect of societal influences, this model has been extensively used in marketing to forecast consumer actions, including purchase decisions. Though the TPB offers a useful framework for comprehending consumer intents, it is sometimes inadequate in its capacity to forecast behavior in complicated, real-world contexts where several elements interact dynamically. According to the TPB, buying behavior is influenced by customer intentions, which are formed by attitudes, subjective norms, and perceived behavioral control. This concept has been used successfully in many other settings, including green consumer behavior, where ecologically friendly purchases heavily depend on intention [54]. Research shows that TPB components—especially perceived behavioral control—are crucial in both online and offline buying decisions, thereby stressing the relevance of the model in the digital era [55].

The Consumer Choice-Making Process (CDMP) describes customers' phases when deciding what to buy. It is another important model, including problem detection, information search, the appraisal of options, purchase choice, and post-purchase behavior. This process-oriented approach emphasizes the sequential character of consumer decision making and has been very helpful in establishing strategies for impacting consumers at several phases of their paths [56]. Like the TPB, the CDMP model needs to improve in capturing the subtleties of consumer behavior, especially in digital environments where decisions are sometimes shaped by real-time data and fast-changing surroundings [57–59].

Generative AI models present a chance to combine and expand these conventional beliefs by allowing for more detailed knowledge of customer behavior to be obtained. For instance, VAEs can provide insights that transcend the explicit elements considered in TPB or CDMP by simulating the latent processes underneath customer decisions. This helps companies forecast more precisely what consumers will do and why they will do it, supporting more focused and successful initiatives [60,61]. By simulating the dynamic interaction between several phases of the consumer journey and how these stages are impacted by outside events, such as marketing messages or social media interactions, transformers—with their capacity to process sequential data—can help us to grasp the decision-making process [62].

The consumer decision-making process consists of several phases, from problem recognition to post-purchase analysis. Well-known models, such as those by Engel, Kollat, and Blackwell, offer insights into consumer interaction with marketing stimuli and choice. These models have evolved to include modern elements, such as digitalization and changing consumer needs, highlighting their relevance in creating successful marketing plans [63]. These models show a need for continuous improvement and integration with

other theories as, although they provide insightful analysis, they also have limits in fully explaining impulsive behaviors or environmental influences [54].

2.4. Generative AI Applications in Consumer Insights and Marketing

Using generative AI in marketing has created fresh opportunities for knowledge and influences consumer behavior. These technologies let companies produce highly customized marketing plans, better evaluate vast customer data, and obtain fresh insights.

Personalized marketing is among the most important fields which generative AI has impacted. Traditional personalizing methods often depend on rule-based algorithms that classify users into established groups. By analyzing enormous volumes of behavioral data and spotting minor trends that might not be clear-cut using conventional approaches, generative AI can, on the other hand, produce unique customer profiles. GANs have been used, for example, to replicate how consumers with particular traits could react to various marketing stimuli, enabling companies to customize their messaging to individual tastes with hitherto unheard-of accuracy [34]. This degree of personalizing can greatly increase consumer engagement and loyalty [64,65].

Consumer behavior forecasting and predictive modeling are two more important uses as well. Transformers let companies examine consumer encounters to forecast future behavior. In e-commerce, for instance, a transformer model might examine a consumer's browsing and purchase behavior to forecast their next likely purchase, allowing the platform to provide relevant and timely product recommendations [62]. Generative AI models can be used in sentiment analysis and emotion recognition, therefore offering a closer understanding of customer attitudes and emotions. Using their interactions with a brand, VAEs, for instance, can replicate consumers' latent emotional states, enabling marketers to identify the fundamental elements influencing customer pleasure or dissatisfaction [46]. This knowledge can then be applied to improve customer service, hone marketing plans, and forge closer consumer bonds.

3. Methodology

A comprehensive search strategy was employed to identify relevant studies examining the application of generative AI techniques in predicting consumer behavior. The search was conducted using Scopus with the following refined queries (titles, abstracts, and keywords):

- ("Generative AI" OR "Generative Adversarial Network" OR "GAN" OR "Variational Autoencoder" OR "VAE" OR "Transformer" OR "AI-driven" OR "Deep Learning" OR "Neural Networks");
- (Title, abstract, and keywords: "Consumer behavior" OR "Customer behavior" OR "Consumer prediction" OR "Customer prediction" OR "Consumer insights" OR "Customer insights" OR "Behavioral prediction" OR "Shopping behavior" OR "Purchase intent").

The number of studies searched from Scopus is depicted in the PRISMA flowchart (Figure 3) [66]. The search was limited to English articles published between 2018 and 2024, focusing on empirical results, case studies, or reviews on the effectiveness and applications of generative AI models in consumer behavior prediction. Initially, 150 documents were found, of which 60 met the inclusion criteria after preliminary screening. A final selection of 31 studies was made based on a detailed review of the titles and abstracts.

Several limitations are acknowledged in this methodology. Potential biases include publication bias, as the search strategy may overlook unpublished or less accessible studies, and language bias due to the exclusion of non-English articles. Limitations of the search strategy may involve incomplete coverage of relevant studies in the literature or variability in data quality across studies.

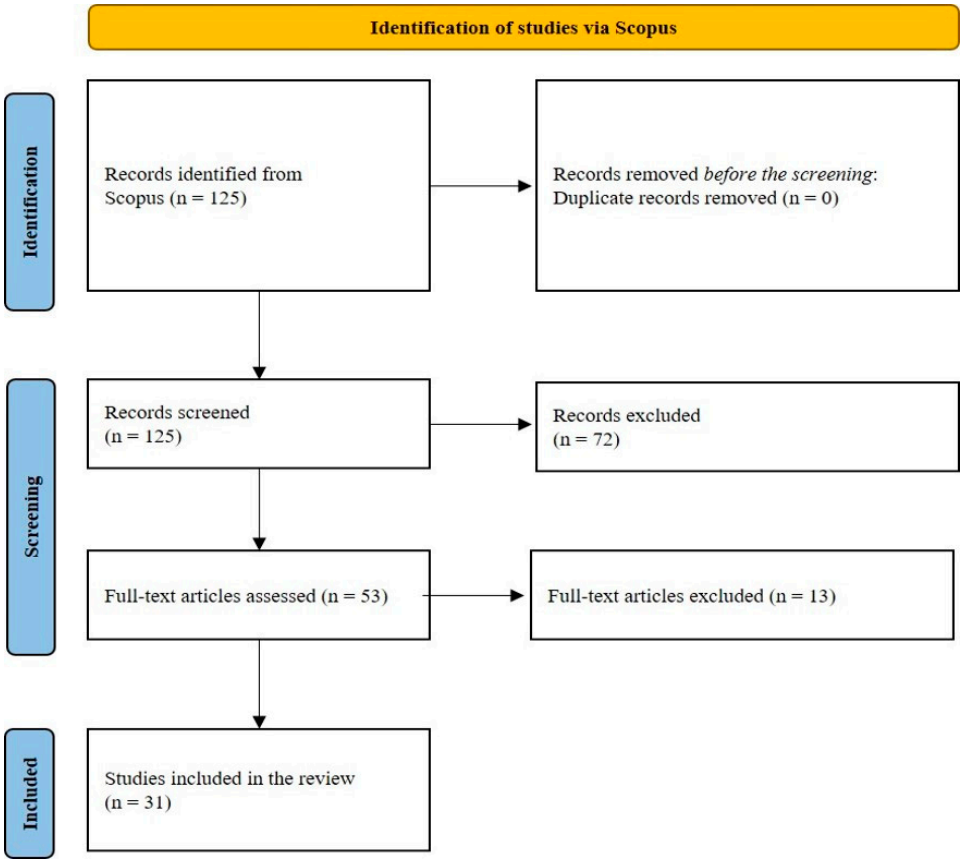


Figure 3. A flow diagram of the study selection process.

4. Findings

4.1. AI in Customer Insights

In recent years, AI has significantly changed how companies collect and evaluate consumer data. This has positively impacted several industries, including public health, transportation, and e-commerce. Deep learning and NLP technologies have made it possible to analyze consumer behavior from massive datasets with more complex and nuanced data, which has been a major driving force behind this progress.

A noteworthy example of AI's application to customer insights is Ha et al.'s research on consumer experiences with electric vehicles (EVs) [65,67–70]. The researchers analyzed the unstructured text from user reviews of EV charging stations using transformer-based deep learning models. Transformers—well known for their efficiency when processing sequential data—were employed to divide reviews into many groups, with the classification accuracy surpassing 91%. This method not only performed better than earlier algorithms, but it also offered insightful real-time data on customer experiences. Policymakers can address locations where users may be underserved by EV infrastructure decisions made using these findings. This application highlights how AI may improve infrastructure planning and public policy by utilizing hitherto unused data sources.

Similarly, Raja and Prema [68] used a hybrid model that combined support vector machines (SVMs) with GPT to investigate the relationship between AI and consumer engagement. Compared to conventional algorithms, their model's classification and clustering accuracy in analyzing textual and sentimental consumer activities was significantly higher. With this hybrid strategy, businesses better understand customer preferences and habits, which helps them customize their strategies and increase engagement. The study demonstrates how combining various AI techniques can result in more successful consumer segmentation and targeted marketing tactics by merging SVM and transformer models.

To study shopping behavior, Liu et al. [69] developed the VATA model, a deep learning framework that integrates VAE, transformers, and attention processes. The VATA model provides a more precise and insightful analysis of consumer trends by effectively capturing the implicit aspects of user behavior and comprehending intricate buying scenarios. With improved buying pattern prediction, this model shows how cutting-edge AI approaches may be used in e-commerce to improve consumer satisfaction and service experiences. Peng et al. [70] used a transformer-based choice model to simulate consumer purchase behavior and assess algorithmic choices in assortment optimization. The methodology creates artificial datasets that mimic consumer behavior, offering a reliable means of evaluating assortment tactics apart from real-world data. This method shows how AI may improve inventory control and revenue optimization by enabling firms to investigate the efficacy of different decision algorithms in optimizing product assortments.

According to Zhang et al. [71], aspect-based sentiment analysis (ABSA) is another example of how AI may improve customer insights. To increase the precision of sentiment analysis in user evaluations, the researchers suggested a deep-level sentiment annotation system utilizing BERT (Bidirectional et al. from Transformers). This methodology provides a more thorough understanding of customer preferences and emotions by addressing the difficulties associated with data annotation and sentiment categorization. These sophisticated sentiment analysis technologies are essential for companies that want to measure consumer happiness and adjust their products and services appropriately. Wang et al.'s [72] demonstration of the application of AI to gather consumer insights extends to public health. In order to understand public mood and communication needs, their study used topic modeling and emotion analysis with NLP approaches to examine tweets about COVID-19 and vaccines. This study demonstrates how AI can draw insightful conclusions from vast amounts of unstructured social media data to inform public health policies.

With cutting-edge techniques like the Mannequin2Real framework created by Zhang et al. [73], the fashion industry benefits from AI. Mannequin photos are converted into photorealistic models using this two-stage generation framework, improving the online buying experience without the expensive expenditures associated with traditional modeling. Similarly, Sohn et al. [74] investigated how customers reacted to fashion products created using GANs and found that they were more eager to pay for AI-designed goods. These studies show how AI may improve product presentation and the comprehension of consumer values to spur innovation in sectors that interact with consumers. Table 2 presents an overview of recent research employing generative AI models across various applications in consumer insights.

Table 2. Summary of studies utilizing generative AI models for consumer insights and predictions.

Study	Generative AI Model	Findings	Year	Cited by
[67]	Transformer-based deep learning	Achieved >91% classification accuracy in analyzing EV charging station reviews; informs public policy and infrastructure planning	2021	21
[68]	Hybrid SVM-GPT transformer model	Hybrid model improves accuracy in classifying and clustering consumer data and aids in customer segmentation and targeted marketing	2024	0
[69]	VATA model (VAE, transformer, attention mechanism)	Captures implicit characteristics of user behavior; enhances analysis of complex shopping scenarios in e-commerce	2024	5
[70]	Transformer-based choice model	Generates synthetic datasets to simulate consumer purchasing behavior; evaluates effectiveness of decision algorithms in inventory management	2024	1

Table 2. *Cont.*

Study	Generative AI Model	Findings	Year	Cited by
[71]	BERT for ABSA	Improves sentiment classification accuracy; provides detailed understanding of consumer preferences and emotions	2021	31
[72]	NLP techniques (LDA, BERT)	Analyzes COVID-19-related tweets to understand public sentiment; offers insights for effective health communication strategies	2023	2
[73]	Mannequin2Real framework (generative model)	Transforms mannequin images into photorealistic model images; enhances online shopping experience by reducing modeling costs	2024	0
[74]	GANs	Higher willingness to pay for GAN-generated fashion products; use of GANs enhances perceived value and consumers' willingness to pay	2021	38

4.2. Generative AI for Consumer Behavior Prediction

In the field of predicting consumer behavior, generative AI has become a game-changer, especially when it comes to e-commerce platforms. This method uses deep learning developments and complex algorithms to improve personalized recommendations, maximize sales tactics, and forecast consumer behavior. Mamta and Sangwan's [75] ensemble deep learning-based prediction framework, AaPiDL, is one of the most well-known frameworks in this discipline. The AaPiDL framework combines convolutional neural networks (CNN), GAN, and long short-term memory RNN (LSTM-RNN) to assess and forecast consumer behavior. Because it uses two modules for purchasing intention and abandonment analysis, this model excels at managing the complexity of customer interactions on e-commerce platforms. A more sophisticated comprehension of client preferences is made possible by the ensemble technique, and this is essential for improving recommender system performance and user experience in general. The framework shows enhanced accuracy and generalization capabilities by assessing performance using metrics like precision, recall, accuracy, and F1-score, opening the door for more successful and customized consumer suggestions.

Similarly, Wei et al. [76] proposed a Double-GAN model to address the problem of scarce consumer behavior data on e-commerce platforms. This approach integrates data from e-commerce platforms through iterative compensation to overcome data sparsity. The JUST model even improves user behavior representation by incorporating structural and semantic information. This methodology is especially helpful for increasing the accuracy of behavioral predictions and coordinating user accounts across several platforms. The study's UBC2vec approach incorporates user interests and improves random walk algorithms to describe the data feature space better. The work efficiently lowers the computational cost of user alignment by creating a "user-commodity" bipartite graph, showing that sophisticated generative techniques can greatly enhance data handling and predictive accuracy in e-commerce contexts.

Hasumoto and Goto [77] have made a noteworthy addition to consumer behavior prediction. Their research centers on utilizing VAEs to forecast client attrition. The work improves churn prediction models by utilizing latent factors from purchase histories, especially in platform firms with more complex customer behavior. The method delivers significant increases in prediction performance, such as a 1.5% rise in the F-measure and a 20% improvement for consumers with recent transactions, by integrating a VAE with real customer distributions. The method provides useful insights for companies trying to reduce customer attrition and improve customer retention by demonstrating the effectiveness of contemporary deep learning techniques in identifying probable churners and developing focused retention measures.

Mena et al. [78] investigated using time-varying recency, frequency, and monetary (RFM) metrics to apply DNNs to customer churn prediction. When focused on time-varying RFM characteristics, RNNs perform better than transformers according to research evaluating their respective performances. This work emphasizes the importance of selecting the appropriate deep-learning architecture for churn prediction. It shows that hybrid approaches—which combine DNN outputs with traditional models—do not always result in better performance. The results support the stand-alone application of DNNs, especially RNNs, for modeling sequential customer behavior data and churn prediction, which leads to more precise and useful customer insights. Table 3 lists works on generative AI models for consumer behavior prediction.

Table 3. Overview of generative AI models for consumer behavior prediction.

Study	Generative AI Model	Findings	Year	Cited by
[75]	GAN, CNN, LSTM-RNN	Proposed ensemble framework improving personalized recommendations and accuracy in e-commerce sales predictions	2024	1
[76]	Double-GAN	Developed model to address data sparsity and computational complexity, improving user alignment across platforms	2023	2
[77]	VAE	Extracted latent features for improved churn prediction, showing 1.5% improvement in F-measure	2022	9
[78]	RNN, Transformers	RNNs outperformed transformers in churn prediction using time-varying features, with hybrid models showing no improvement	2024	5
[79]	LLMs	Highlighted effectiveness of document retrieval techniques for dynamic marketing insights	2024	0
[80]	Generative AI (Convergent and Divergent Thinking)	Analyzed AI's impact on consumer behavior, focusing on predictive and generative AI advancements	2024	2
[81]	Multivariate Behavior Sequence Transformer (MBST)	Achieved high F-score and AUC in churn prediction with novel transformer-based model	2022	11
[82]	Conditional GAN (CTGAN), SMOTE, Ensemble Models	Used class balancing techniques and hybrid ensemble models to improve churn prediction accuracy	2024	0
[83]	TN-GAN, CNN, LSTM	Improved behavioral prediction using TN-GAN and hybrid CNN-LSTM model for pet monitoring systems	2023	2
[84]	CNN, LSTM	Enhanced priority detection in neuromarketing using EEG features and machine learning	2024	0
[85]	Causal VAEs, Graph Neural Networks	Developed explainable model with high accuracy in predicting e-commerce purchase behavior	2024	0
[86]	M-GAN-XGBOOST	Proposed model combining LSTM, GAN, and XGBOOST for accurate sales predictions and marketing strategies	2021	22
[87]	Generative AI	Evaluated EV charging impact on grids, finding that proper siting can handle 100% EV adoption without upgrades	2024	0
[88]	Battery Storage Systems	Explored battery storage solutions based on consumer energy behavior to improve grid operability	2022	0

Table 3. *Cont.*

Study	Generative AI Model	Findings	Year	Cited by
[89]	Distributed Energy Resources Management	Proposed system optimizing power dispatch, leading to significant cost savings and improved power quality	2021	9
[90]	Load Management Methodology	Demonstrated improved load management and system stability through new methodology	2020	10
[91]	Orthogonal Particle Swarm Optimization	Developed model for load demand forecasting and transformer burden analysis in Baghdad City	2020	5
[92]	Generative AI	Assessed impact of low-carbon technologies on grid infrastructure, highlighting spatial and temporal variations	2023	4

4.3. Generative AI for Energy Data Modeling

In energy data modeling, generative AI—particularly through methods like GANs—has become a game-changer, providing creative answers to some of the industry's most persistent problems. This topic explores the use of generative AI for micro-grid energy resource management, non-intrusive load monitoring (NILM) model optimization, and the creation and improvement of building electrical load profiles. We may gain a deeper understanding of how these technologies influence consumer behavior analysis and energy management by looking at current developments and approaches (Table 4).

Table 4. Overview of studies on generative AI models for energy data applications.

Study	Generative AI Model	Findings	Year	Cited by
[93]	GAN	GAN approach captures real load profile patterns and maintains data privacy	2020	64
[94]	BERT-LSTM, CNN	Improves classification accuracy (0.957) and power disaggregation	2024	0
[95]	Coupled GANs (C-GAN)	Enhanced NILM framework saves energy costs and boosts market revenue	2023	10
[96]	VAE, Temporal Convolutional Networks	Two-stage model improves forecasting accuracy (16% MAE; 19% MAPE)	2023	10
[97]	Multi-Objective Optimization Model	HEMS reduces costs (31%), demand peak (18%), and transformer Loss of Life (28%)	2021	35

The work of Wang and Hong [93] marks a substantial advancement in the creation of realistic building electrical load profiles through the use of GANs. Conventional techniques for creating these profiles are frequently labor-intensive and need to catch up in capturing the dynamic, random character of actual buildings. GANs are used in Wang and Hong's method to overcome these drawbacks. Their process entails normalizing daily load profiles, applying GANs to produce realistic load profiles for each cluster, and then grouping them using the k-means algorithm. The Building Data Genome Project was used to validate this method, and the findings demonstrate that the created profiles closely matched the statistical characteristics of the actual profiles. The KL divergence metric proved that the GAN-based approach was effective by showing that there is little difference between the produced and real load profiles. This development improves load profile creation accuracy and offers a way to anonymize data from smart meters, resolving privacy issues and facilitating research and grid-interactive applications.

Similarly, Xu et al. [94] created an intelligent NILM model that combines power encoding with convolutional state modules to enhance load disaggregation. The significance of precisely separating on/off states is frequently overlooked by traditional NILM techniques,

which can result in considerable mistakes in power consumption estimates. To bridge this gap, Xu et al. adopted a dual-module system: a convolutional state module (CSM) that concentrates on disaggregating device states and a power encoding module that uses long short-term memory networks and BERT-LSTM for initial energy disaggregation. The total accuracy is improved by the more accurate disaggregation made possible by integrating these modules. The model outperformed baseline models regarding device state classification and error reduction, indicating its potential to provide insights into energy usage and support reduction tactics.

By putting forth a paradigm for micro-grid energy management that includes Coupled GANs (C-GANs), Tao et al. [95] expanded the application of NILM even further. This research tackles the difficulty of acquiring comprehensive micro-grid appliance data because of technological and privacy limitations. Tao et al. created a system that assesses HVAC unit reserve capacity without requiring comprehensive appliance data by utilizing NILM and C-GANs. Their methodology entails utilizing Hidden Markov Models (HMMs) to evaluate reserve capacity and using NILM to break down electricity utilization. This approach helps to optimize market bidding methods and enhances the accuracy of reserve evaluations, which ultimately benefits market operators by reducing costs and boosting income.

A new method for short-term household load forecasting (STLF) based on VAEs' deep generative model is presented by Langevin et al. [96]. This approach uses past and future appliance load data to overcome the shortcomings of conventional forecasting techniques. The two-stage method uses NILM to estimate granular load data from aggregate observations and then forecasts future loads using a predictive model. This approach takes into account the fact that household energy usage is dynamic, which improves forecasting accuracy. The outcomes highlight the potential of deep generative models in improving energy management systems by showing notable improvements in the mean absolute error (MAE) and mean absolute percentage error (MAPE) compared to state-of-the-art techniques.

5. Discussion

5.1. Thematic Analysis

The subject of consumer behavior prediction has greatly advanced because of generative AI approaches such as transformers, VAEs, and GANs. The main topics and trends that have emerged in recent research on the use of generative AI in consumer behavior prediction and comprehension are examined in this thematic analysis. The emphasis is on how these technologies are transforming different industries, how different AI models might be integrated, and how consumer insights are changing.

The significant impact of transformers on the forecast of consumer behavior is a major theme. Transformers have gained widespread use because of their efficaciousness in deciphering unstructured data. They are recognized for their capacity to manage sequential and textual data. Transformers, for example, have been shown through studies to assess user evaluations with remarkable accuracy, with classification rates of 91% [67]. This capacity makes gaining complex insights into customer experiences easier, which is essential for applications like policymaking and infrastructure planning. Through more accurate predictions, businesses can improve customer engagement and refine their marketing strategies through the use of hybrid models that combine transformers with techniques like SVM and show enhanced accuracy in classifying and clustering customer data [68].

Another important theme is applying hybrid AI models, which combine several AI methods to provide better performance. VAE–transformer–attention mechanism models are combined to offer a more thorough explanation of complicated consumer behaviors. This integration makes better analysis and forecasting of consumer trends and shopping behaviors possible [69]. By combining data from several sources to overcome data sparsity, hybrid models have improved behavioral predictions and synchronized user accounts across platforms [76]. These models create more reliable and accurate forecasting frameworks by utilizing the advantages of several AI techniques.

The impact of generative AI on consumer insights and marketing is another important issue. Cutting-edge AI methods are changing marketing tactics by providing more insightful, useful information about customer behavior. Transformer-based models, for instance, have been used to analyze consumer purchasing behavior and offer tools for inventory management and assortment optimization [70]. Businesses can use this capability to test and improve their strategies by employing artificial datasets that replicate the behaviors of real consumers. A more in-depth understanding of customer preferences and feelings is provided by sophisticated sentiment analysis approaches, such as ABSA employing models like BERT [71]. This degree of detail is necessary for customizing goods and services to satisfy unique client requirements.

The transformational potential of generative AI is highlighted by its wide range of applications across multiple fields. When NLP approaches are used on social media data, they can provide important insights into public attitudes and the communication needs around health crises in the field of public health [72]. This application highlights how AI may extract relevant data from big databases, which helps inform public health efforts. The fashion sector has also improved the online buying experience using AI advances. For instance, frameworks that create photorealistic models from mannequin photos enhance the way products are presented while also cutting the expense of using more conventional modeling techniques [73]. This development illustrates how AI may spur innovation and raise consumer value in sectors where consumers are involved.

5.2. Comparative Analysis

According to a comparative investigation, the usefulness, efficiency, and application of generative AI models in forecasting consumer behavior vary significantly depending on the environment. The main models being examined are transformers, VAEs, and GANs, each with advantages and disadvantages when predicting customer behavior.

GANs have demonstrated impressive results in mimicking customer behavior and producing synthetic data. Their competitive training of a generator and a discriminator yields realistic data distributions, which is their main strength. For example, Wei et al.'s [76] Double-GAN model integrates data from disparate e-commerce platforms to solve data sparsity concerns effectively. This method improves prediction accuracy by capturing intricate user behavior patterns through iterative data alignment and compensation. When it is possible to provide high-quality synthetic data to enhance prediction models, GANs perform very well. Nevertheless, mode collapse—a situation in which the generator generates few data variations—and high processing costs during training are problems that GANs must also overcome. These restrictions may affect how well the approach scales and applies to various consumer segments.

Another method is to utilize VAEs, which can capture complex distributions and provide a variety of data samples by learning probabilistic mapping from data to a latent space. VAEs have been used to predict client attrition, as Hasumoto and Goto [77] have shown. Their work demonstrates how adding latent factors from purchase histories to VAEs improves churn prediction models. When reflecting the underlying probabilistic character of consumer behavior is critical, VAEs perform exceptionally well. Their strength is in offering a latent space that is smooth and suitable for a wide range of generative operations. Nonetheless, VAEs may not have generative powers as strong as those of GANs because of the intrinsic trade-off between latent space regularization and reconstruction accuracy, which results in less realistic reconstructions.

Transformers' self-attention techniques enable efficient modeling of long-range dependencies, constituting a substantial leap in managing textual and sequential data. Transformers are better at capturing subtle patterns in big datasets when applied to consumer review and engagement analysis, as demonstrated by the research conducted by Ha et al. [67] and Raja and Prema [68]. Transformers with excellent classification and clustering accuracy, such as BERT and GPT, are perfect for sentiment analysis applications and complicated NLP. They are effective tools for enhancing marketing campaigns and client interaction due

to their ability to manage massive amounts of data and deliver real-time insights. However, transformers can be a hurdle for smaller businesses or applications with limited resources because they frequently require complicated fine-tuning and large computational resources.

It is clear from comparing these models that the application's particular needs determine which AI technique is best. While GANs are useful for handling data sparsity and producing synthetic data, training stability could be improved. Although VAEs provide strong probabilistic modeling, they might not be able to produce high-fidelity samples. Transformers are quite accurate and easily interpreted for processing and analyzing text data, but they require much processing power.

Every model has advantages and disadvantages; depending on the situation, each one's usefulness may differ. GANs, for example, might be better suited for jobs involving synthetic data creation. In contrast, transformers might be better suited for applications needing real-time insights and complex language interpretation. The model selected should align with the particular objectives of predicting consumer behavior, considering variables like the type of predictive tasks, computational capacity, and data quality. A well-rounded solution might also be provided by combining different models or creating hybrid strategies, which would take advantage of each method's advantages to overcome drawbacks and improve overall prediction performance.

5.3. Challenges and Limitations

Although promising, generative AI models have drawbacks and restrictions that affect their usefulness and applicability in various fields. The availability and quality of data present one of the main obstacles. Generative AI models like VAEs and GANs frequently need enormous volumes of high-quality labeled data to train efficiently. Acquiring such data can be difficult in many real-world applications, particularly in developing industries like public health and e-commerce [98]. Incomplete, noisy, or biased data might result in poor model performance and erroneous predictions. For example, the generative model may not capture the underlying patterns of consumer preferences if the consumer behavior data are sparse or contains high noise. This could result in less successful marketing strategies or poorly informed corporate decisions. Furthermore, data acquisition can be made more challenging by privacy concerns surrounding gathering and using personal data, making it challenging to train models with large datasets [99].

Computational complexity is a big additional challenge. Advanced generative models—especially those that use deep learning techniques—demand a significant amount of computer power for training and deployment. Training complex models, such as transformers or GANs, requires specialized hardware, such as GPUs or TPUs, and can be computationally demanding [100]. Research groups with low resources or smaller organizations may struggle to handle this computing load. Furthermore, because running large-scale models requires much energy, the high resource requirements may also result in higher operational costs and adverse environmental effects. Some ways to approach this difficulty are using more effective hardware and cloud computing technologies or refining algorithms to fulfill lower processing demands [101].

Generalizability is yet another important constraint. It may be difficult for generative AI models trained on certain datasets or specialized applications to generalize to new or unknown environments. For instance, a model developed to forecast consumer behavior in one market niche might not function effectively when used to anticipate behavior in a different niche with unique features. This generalizability problem is especially important in sectors that change quickly, like consumer behavior, where customer preferences and market dynamics change swiftly. It might take many resources to continuously update and retrain models so they are accurate and useful in various scenarios [102].

There are serious issues with the generative AI models' interpretability and openness. Many sophisticated AI models, such as generative models based on deep learning, function as "black boxes", meaning that humans cannot readily understand how they make decisions. The models' inability to be transparent may make it more difficult for stakeholders to trust

and make good use of them. The necessity for interpretable and transparent models is especially vital in critical applications like healthcare and finance, where decisions based on model outputs might have significant consequences.

5.4. Implications for Practice, Research, and Future Agenda

Generative AI techniques all significantly impact the fields of marketing, advertising, and customer relationship management. Transformer models and hybrid strategies can significantly improve personalized marketing campaigns, which let companies create highly targeted ads based on sentiment research and sophisticated consumer behavior analysis. Recommendation systems are further advanced by GANs and VAEs, which provide instruments for more precise product recommendations and enhance consumer involvement, both of which lead to higher sales and customer happiness. Furthermore, by simulating multiple scenarios and optimizing inventory management without merely depending on past data, transformer-based models for assortment optimization help businesses improve stock management and better respond to changes in the market. VAEs and deep learning models in customer relationship management provide useful insights into customer turnover trends, allowing companies to identify at-risk clients and implement efficient retention measures to lower attrition rates. A deeper understanding of consumer preferences and feedback is made possible by AI-driven sentiment analysis tools, such as those that use BERT. This results in more informed decisions and improved customer experiences.

The literature now has some noticeable gaps, notwithstanding the achievements presented. Numerous studies concentrate on controlled environments or theoretical models, emphasizing the need for more studies evaluating the practicality and scalability of generative AI applications. To comprehend the true performance and difficulties of these models in various contexts, it is imperative to conduct empirical research through case studies and field trials. Further research is necessary to address data privacy issues, computing capacity, and domain adaptability arising from incorporating generative AI techniques into current systems. Subsequent studies should delve into nascent AI technologies, such as federated and self-supervised learning, to tackle present constraints and enhance the potential of generative AI. AI combined with interdisciplinary subjects like psychology, social sciences, and behavioral economics may provide a more thorough understanding of customer behavior and result in more accurate predictive models. Furthermore, ethical and privacy concerns need to be taken into account. Research should concentrate on creating frameworks for responsible AI use and ensuring that data protection laws are followed to preserve customer confidence and legal compliance.

Ethical and privacy concerns are equally essential and necessitate clear frameworks and guidelines for the responsible deployment of AI in consumer data analysis. Responsible AI use is pivotal in maintaining consumer trust, safeguarding privacy, and achieving legal compliance with data protection laws. Key guidelines include adopting data minimization practices, where only essential data are collected and processed, and employing secure data handling practices, such as encryption and anonymization [103]. To mitigate algorithmic bias, organizations should implement diversity-aware training data collection and conduct routine bias audits to identify and address any biased patterns that could influence model outputs. Additionally, creating multi-stakeholder review boards can enhance fairness by offering perspectives from diverse backgrounds, ensuring AI tools do not inadvertently perpetuate stereotypes or exclude specific groups [104].

Ensuring transparency is another cornerstone of ethical AI in consumer data prediction. Clear documentation of AI model designs, decision processes, and data sources enables greater scrutiny and trust. Providing consumers with understandable explanations of how their data are used and the AI's decision-making process is essential for transparency. Furthermore, organizations should consider adopting explainable AI (XAI) techniques to clarify complex model behaviors, thereby fostering trust and understanding among both users and stakeholders. Establishing these frameworks for ethical AI use not only

aligns with legal and regulatory requirements but also enhances customer confidence and organizational accountability [105].

5.5. Proposed Framework

Based on the literature review, we proposed a new conceptual framework entitled AI-Driven Consumer Insight Framework. In the AI-Driven Consumer Insight Framework, businesses begin by systematically gathering and refining consumer data to ensure their quality and readiness for analysis. This process requires tapping into diverse sources, such as social media interactions, transaction records, browsing behaviors, and direct customer feedback. Together, these sources offer a comprehensive perspective on consumers' preferences and actions. Preprocessing is essential to clean, organize, and structure the raw data, transforming them into a consistent and analyzable format. By leveraging AI-driven methods to streamline these tasks, organizations lay the groundwork for accurate consumer behavior analysis.

Next, businesses should select and customize AI models according to specific business objectives and data attributes. Each model type serves unique purposes: Generative Adversarial Networks (GANs) are highly effective for generating synthetic data in cases where real-world samples are limited or need augmentation; transformers are ideal for sequential data analysis, excelling at capturing patterns in data such as purchase histories; and Variational Autoencoders (VAEs) provide insights into hidden patterns by encoding data into a latent space, where underlying trends become more observable. Customizing these models to suit industry-specific needs, such as accounting for regional cultural preferences or seasonal retail trends, further refines their analytical accuracy and relevance.

Deploying AI models within consumer-facing systems allows for real-time consumer behavior insights, a critical asset in today's fast-paced markets. By integrating AI into digital platforms, businesses can continuously analyze incoming consumer data, facilitating real-time behavioral predictions and sentiment analysis. With the ability to monitor interactions instantaneously, organizations can respond dynamically to consumer needs, offering timely and personalized experiences. This agility strengthens customer engagement, aligns marketing efforts with current consumer expectations, and enhances an organization's competitive advantage.

Throughout this process, privacy and ethical considerations are paramount. Organizations must prioritize data privacy and ethical AI practices to maintain consumer trust and regulatory compliance. Techniques such as data anonymization safeguard consumer identities, while privacy-preserving AI methods, like federated learning, enable analysis without direct data sharing. Additionally, businesses should adopt tools to regularly check for and mitigate biases in AI model outputs, ensuring that consumer insights are fair, accurate, and ethical. Meeting data protection regulations, such as GDPR, further supports consumer trust and mitigates legal risks.

Finally, the insights generated through this framework must be actionable to maximize their impact. AI-driven insights tailored to specific consumer segments can guide marketing strategies, product development, and customer engagement. A continuous feedback loop in the AI analytics system allows the model to refine its insights as consumer trends evolve, ensuring strategies remain relevant and effective. This structured approach enables businesses to remain attuned to the latest consumer dynamics, offering an ethically driven, adaptable, and precise understanding of consumer behavior in real time.

6. Conclusions

This systematic review examined the efficacy, limitations, and real-world applicability of GANs, VAEs, and transformers in consumer behavior prediction. The evaluation of 31 studies in e-commerce, energy data modeling, and public health showed that AI models can improve targeted marketing, inventory management, and customer retention. Transformer models excel in handling complicated sequential data for real-time consumer insights, while GANs and VAEs offer powerful solutions for simulating realistic data and

predicting customer behaviors such as churn and purchasing intent. However, the analysis highlights important difficulties, including data protection, integration into existing systems, and limited real-world testing.

Generative AI has the potential to revolutionize consumer behavior prediction. These powerful techniques allow organizations to understand consumer preferences and behaviors with unparalleled accuracy and depth as they increasingly adopt data-driven decision making. Generative AI can synthesize realistic data and extract insights from vast amounts of unstructured data, making it essential for targeted marketing, improving customer experience, and optimizing operations.

To fully realize the potential of generative AI, it is crucial to address current restrictions and effectively integrate new AI technologies. Future research should explore the integration of generative AI models with traditional forecasting methods to enhance accuracy and adaptability in various sectors. Additionally, there is a pressing need to examine frameworks for responsible AI use that prioritize ethical considerations and data privacy, ensuring compliance with regulations while leveraging AI capabilities.

Furthermore, studies focusing on the real-world implementations of generative AI in different industries can provide valuable insights into their effectiveness and applicability. Promoting interdisciplinary collaborations will also foster innovation, enabling the development of comprehensive strategies for integrating AI into consumer behavior prediction. Ultimately, exploring how generative AI can adapt to the ever-changing digital landscape will be essential in addressing emerging consumer trends and preferences. By addressing these challenges and research areas, we can advance the field and ensure that generative AI continues to effectively predict consumer behavior in a rapidly evolving environment.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Lim, W.M.; Rasul, T.; Kumar, S.; Ala, M. Past, present, and future of customer engagement. *J. Bus. Res.* **2022**, *140*, 439–458. [CrossRef]
2. Ciasullo, M.V.; Lim, W.M.; Manesh, M.F.; Palumbo, R. The patient as a prosumer of healthcare: Insights from a bibliometric-interpretive review. *J. Health Organ. Manag.* **2022**, *36*, 133–157. [CrossRef] [PubMed]
3. Chandra, S.; Verma, S.; Lim, W.M.; Kumar, S.; Donthu, N. Personalization in personalized marketing: Trends and ways forward. *Psychol. Mark.* **2022**, *39*, 1529–1562. [CrossRef]
4. Davenport, T.; Guha, A.; Grewal, D.; Bressgott, T. How artificial intelligence will change the future of marketing. *J. Acad. Mark. Sci.* **2020**, *48*, 24–42. [CrossRef]
5. Mahoney, L.M.; Tang, T. *Strategic Social Media: From Marketing to Social Change*; John Wiley & Sons: Hoboken, NJ, USA, 2024.
6. Nosi, C.; Zollo, L.; Rialti, R.; Ciappei, C. Sustainable consumption in organic food buying behavior: The case of quinoa. *Br. Food J.* **2020**, *122*, 976–994. [CrossRef]
7. Avotra, A.A.R.N.; Chenyun, Y.; Yongmin, W.; Lijuan, Z.; Nawaz, A. Conceptualizing the state of the art of corporate social responsibility (CSR) in green construction and its nexus to sustainable development. *Front. Environ. Sci.* **2021**, *9*, 774822. [CrossRef]
8. Dias, A.; Sousa, B.; Santos, V.; Ramos, P.; Madeira, A. Wine tourism and sustainability awareness: A consumer behavior perspective. *Sustainability* **2023**, *15*, 5182. [CrossRef]
9. Ramya, N.; Ali, S.M. Factors affecting consumer buying behavior. *Int. J. Appl. Res.* **2016**, *2*, 76–80.
10. Verma, Y.; Singh, M.R. Marketing mix, customer satisfaction and loyalty: An empirical study of telecom sector in Bhutan. *Indian J. Commer. Manag. Stud.* **2017**, *8*, 121–129.
11. Abd Wahab, N.; Hassan, L.F.A.; Shahid, S.A.M.; Maon, S.N. The relationship between marketing mix and customer loyalty in hijab industry: The mediating effect of customer satisfaction. *Procedia Econ. Financ.* **2016**, *37*, 366–371. [CrossRef]
12. Galvano, D.F. Integrating Consumer Behavior Insights into Effective Marketing Strategies. Available online: https://www.researchgate.net/publication/380075292_Integrating_Consumer_Behavior_Insights_into_Effective_Marketing_Strategies (accessed on 24 August 2024).
13. Sama, R. Impact of media advertisements on consumer behaviour. *J. Creat. Commun.* **2019**, *14*, 54–68. [CrossRef]
14. Glickman, M.; Zhang, Y. AI and generative AI for research discovery and summarization. *Harv. Data Sci. Rev.* **2024**, *6*, 1–34. [CrossRef]

15. Taherdoost, H.; Madanchian, M. Artificial intelligence and sentiment analysis: A review in competitive research. *Computers* **2023**, *12*, 37. [CrossRef]
16. Bandi, A.; Adapa, P.V.S.R.; Kuchi, Y.E.V.P.K. The power of generative ai: A review of requirements, models, input–output formats, evaluation metrics, and challenges. *Future Internet* **2023**, *15*, 260. [CrossRef]
17. Gupta, P.; Ding, B.; Guan, C.; Ding, D. Generative AI: A systematic review using topic modelling techniques. *Data Inf. Manag.* **2024**, *8*, 100066. [CrossRef]
18. Zhou, G.; Xie, S.; Hao, G.; Chen, S.; Huang, B.; Xu, X.; Wang, C.; Zhu, L.; Yao, L.; Zhang, K. Emerging synergies in causality and deep generative models: A survey. *arXiv* **2023**, arXiv:2301.12351.
19. Brophy, E.; Wang, Z.; She, Q.; Ward, T. Generative adversarial networks in time series: A systematic literature review. *ACM Comput. Surv.* **2023**, *55*, 1–31. [CrossRef]
20. Cronin, I. *Understanding Generative AI Business Applications*; Springer Nature: Dordrecht, The Netherlands, 2024.
21. Ayemowa, M.O.; Ibrahim, R.; Khan, M.M. Analysis of Recommender System Using Generative Artificial Intelligence: A Systematic Literature Review. *IEEE Access* **2024**, *12*, 87742–87766. [CrossRef]
22. Yu, H.; Guo, Y. Generative artificial intelligence empowers educational reform: Current status, issues, and prospects. In *Frontiers in Education*; Frontiers Media SA: Lausanne, Switzerland, 2023; p. 1183162.
23. Eke, D.O. ChatGPT and the rise of generative AI: Threat to academic integrity? *J. Responsible Technol.* **2023**, *13*, 100060. [CrossRef]
24. Banh, L.; Strobel, G. Generative artificial intelligence. *Electron. Mark.* **2023**, *33*, 63. [CrossRef]
25. Perov, V.; Perova, N. AI Hallucinations: Is "Artificial Evil" Possible? In Proceedings of the 2024 IEEE Ural-Siberian Conference on Biomedical Engineering, Radioelectronics and Information Technology (USBEREIT), Yekaterinburg, Russian, 13–15 May 2024; pp. 114–117.
26. Elahi, M.; Afolaranmi, S.O.; Martinez Lastra, J.L.; Perez Garcia, J.A. A comprehensive literature review of the applications of AI techniques through the lifecycle of industrial equipment. *Discov. Artif. Intell.* **2023**, *3*, 43. [CrossRef]
27. Jain, V.; Wadhwani, K.; Eastman, J.K. Artificial intelligence consumer behavior: A hybrid review and research agenda. *J. Consum. Behav.* **2024**, *23*, 676–697. [CrossRef]
28. Ebrahimi, P.; Basirat, M.; Yousefi, A.; Nekmahmud, M.; Gholampour, A.; Fekete-Farkas, M. Social networks marketing and consumer purchase behavior: The combination of SEM and unsupervised machine learning approaches. *Big Data Cogn. Comput.* **2022**, *6*, 35. [CrossRef]
29. Kliestik, T.; Zvarikova, K.; Lăzăroiu, G. Data-driven machine learning and neural network algorithms in the retailing environment: Consumer engagement, experience, and purchase behaviors. *Econ. Manag. Financ. Mark.* **2022**, *17*, 57–69.
30. Wang, Y.; Yu, C. Social interaction-based consumer decision-making model in social commerce: The role of word of mouth and observational learning. *Int. J. Inf. Manag.* **2017**, *37*, 179–189. [CrossRef]
31. Howard, J.A. *The Theory of Buyer Behavior*; John Willey & Sons: Hoboken, NJ, USA, 1969; Volume 145.
32. Azad, M.S.; Khan, S.S.; Hossain, R.; Rahman, R.; Momen, S. Predictive modeling of consumer purchase behavior on social media: Integrating theory of planned behavior and machine learning for actionable insights. *PLoS ONE* **2023**, *18*, e0296336. [CrossRef]
33. Sharma, P.; Kumar, M.; Sharma, H.K.; Biju, S.M. Generative adversarial networks (GANs): Introduction, Taxonomy, Variants, Limitations, and Applications. *Multimed. Tools Appl.* **2024**, 1–48. [CrossRef]
34. Goodfellow, I.; Pouget-Abadie, J.; Mirza, M.; Xu, B.; Warde-Farley, D.; Ozair, S.; Courville, A.; Bengio, Y. Generative adversarial nets. In *Advances in Neural Information Processing Systems 27*; NeurIPS: San Diego, CA, USA, 2014.
35. Kingma, D.P. Auto-encoding variational bayes. *arXiv* **2013**, arXiv:1312.6114.
36. Babu, S.M.; Kumar, P.P.; Devi, S.; Reddy, K.P.; Satish, M. Predicting Consumer Behaviour with Artificial Intelligence. In Proceedings of the 2023 IEEE 5th International Conference on Cybernetics, Cognition and Machine Learning Applications (ICCCMLA), Hamburg, Germany, 7–8 October 2023; pp. 698–703.
37. Naveeenkumar, N.; Rallapalli, S.; Sasikala, K.; Priya, P.V.; Husain, J.; Boopathi, S. Enhancing Consumer Behavior and Experience Through AI-Driven Insights Optimization. In *AI Impacts in Digital Consumer Behavior*; IGI Global: Hershey, PA, USA, 2024; pp. 1–35.
38. Pasupulati, R.; Mahendran, J.; Majumdar, A. Predictive Analysis of Digital Consumer Behaviour: Key to Understanding Success and Failure of a Brand. In *Enhancing and Predicting Digital Consumer Behavior with AI*; IGI Global: Hershey, PA, USA, 2024; pp. 238–267.
39. Feuerriegel, S.; Hartmann, J.; Janiesch, C.; Zschech, P. Generative ai. *Bus. Inf. Syst. Eng.* **2024**, *66*, 111–126. [CrossRef]
40. Akhtar, Z.B. Unveiling the evolution of generative AI (GAI): A comprehensive and investigative analysis toward LLM models (2021–2024) and beyond. *J. Electr. Syst. Inf. Technol.* **2024**, *11*, 22. [CrossRef]
41. Kalota, F. A Primer on Generative Artificial Intelligence. *Educ. Sci.* **2024**, *14*, 172. [CrossRef]
42. Sun, Q.; Cui, Y.; Zhang, X.; Zhang, F.; Yu, Q.; Wang, Y.; Rao, Y.; Liu, J.; Huang, T.; Wang, X. Generative multimodal models are in-context learners. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, Seattle, WA, USA, 16–22 June 2024; pp. 14398–14409.
43. Meskó, B. The impact of multimodal large language models on health care's future. *J. Med. Internet Res.* **2023**, *25*, e52865. [CrossRef] [PubMed]

44. Wang, J.; Liu, Y. Prediction of Sensitive Consumer Behavior Based on Random Forest with Grid Search Cross-Validation. In Proceedings of the 2023 3rd International Conference on Mobile Networks and Wireless Communications (ICMNWC), Tumkur, India, 4–5 December 2023; pp. 1–4.
45. Prosvetov, A. GAN for recommendation system. *J. Phys. Conf. Ser.* **2019**, *1405*, 012005. [CrossRef]
46. Higgins, I.; Matthey, L.; Pal, A.; Burgess, C.P.; Glorot, X.; Botvinick, M.M.; Mohamed, S.; Lerchner, A. beta-vae: Learning basic visual concepts with a constrained variational framework. *ICLR (Poster)* **2017**, *3*, 1–22.
47. Yoon, J.H.; Jang, B. Evolution of deep learning-based sequential recommender systems: From current trends to new perspectives. *IEEE Access* **2023**, *11*, 54265–54279. [CrossRef]
48. Yenduri, G.; Ramalingam, M.; Selvi, G.C.; Supriya, Y.; Srivastava, G.; Maddikunta, P.K.R.; Raj, G.D.; Jhaveri, R.H.; Prabadevi, B.; Wang, W. Gpt (generative pre-trained transformer)—A comprehensive review on enabling technologies, potential applications, emerging challenges, and future directions. *IEEE Access* **2024**, *12*, 54608–54649. [CrossRef]
49. Gupta, R.; Nair, K.; Mishra, M.; Ibrahim, B.; Bhardwaj, S. Adoption and impacts of generative artificial intelligence: Theoretical underpinnings and research agenda. *Int. J. Inf. Manag. Data Insights* **2024**, *4*, 100232. [CrossRef]
50. Paz, M.D.R.; Vargas, J.C.R. Main theoretical consumer behavioural models. A review from 1935 to 2021. *Heliyon* **2023**, *9*, e13895.
51. Manuere, H.T.; Chikazhe, L.; Manyeruke, J. Theoretical models of consumer behaviour: A literature review. *Int. J. Educ. Humanit. Soc. Sci.* **2022**, *5*, 105–112. [CrossRef]
52. Khegay, E.; Aubakirov, S. Theoretical Exploration of Consumer Behavior. *Eurasian J. Econ. Bus. Stud.* **2021**, *61*, 49–61. [CrossRef]
53. Ajzen, I. The Theory of Planned Behavior. In *Organizational Behavior and Human Decision Processes*; Elsevier: Amsterdam, The Netherlands, 1991.
54. Zulfikar, R.; Suryadi, N.; Prasarry, Y.V.; Barqiah, S. Penggunaan Theory of Planned Behavior dalam Kajian Perilaku Konsumen Hijau. *JKBM (JURNAL KONSEP BISNIS DAN MANAJEMEN)* **2023**, *10*, 28–41. [CrossRef]
55. Kurniawati, M.; Adeline, M.; Ramadhan, K.; Irwansyah, I. Implementation theory of planned behavior on the purchase decision online and offline. *Asian J. Eng. Soc. Health* **2023**, *2*, 1119–1132. [CrossRef]
56. Blackwell, R.D.; Miniard, P.W.; Engel, J.F. *Consumer Behavior Thomson*; South-Western: Mason, OH, USA, 2006.
57. Ajzen, I. Consumer attitudes and behavior: The theory of planned behavior applied to food consumption decisions. *Ital. Rev. Agric. Econ.* **2015**, *70*, 121–138.
58. Rozenkowska, K. Theory of planned behavior in consumer behavior research: A systematic literature review. *Int. J. Consum. Stud.* **2023**, *47*, 2670–2700. [CrossRef]
59. Islam, Q.; Ali Khan, S.M.F. Assessing consumer behavior in sustainable product markets: A structural equation modeling approach with partial least squares analysis. *Sustainability* **2024**, *16*, 3400. [CrossRef]
60. Mariani, M.; Dwivedi, Y.K. Generative artificial intelligence in innovation management: A preview of future research developments. *J. Bus. Res.* **2024**, *175*, 114542. [CrossRef]
61. Mancisidor, R.A.; Kampffmeyer, M.; Aas, K.; Jenssen, R. Learning latent representations of bank customers with the variational autoencoder. *Expert Syst. Appl.* **2021**, *164*, 114020. [CrossRef]
62. Sun, F.-Y.; Hoffmann, J.; Verma, V.; Tang, J. Infograph: Unsupervised and semi-supervised graph-level representation learning via mutual information maximization. *arXiv* **2019**, arXiv:1908.01000.
63. Makhrout, S.; Chouhbi, A. Fundamental Models of Consumer Purchasing Behavior: An In-Depth Analysis Since the 1960s. *[RMd] Rev.* **2024**, *6*, e202419.
64. Gavilanes, J.M.; Flatten, T.C.; Brettel, M. Content strategies for digital consumer engagement in social networks: Why advertising is an antecedent of engagement. *J. Advert.* **2018**, *47*, 4–23. [CrossRef]
65. Harmeling, C.M.; Moffett, J.W.; Arnold, M.J.; Carlson, B.D. Toward a theory of customer engagement marketing. *J. Acad. Mark. Sci.* **2017**, *45*, 312–335. [CrossRef]
66. Page, M.J.; McKenzie, J.E.; Bossuyt, P.M.; Boutron, I.; Hoffmann, T.C.; Mulrow, C.D.; Shamseer, L.; Tetzlaff, J.M.; Akl, E.A.; Brennan, S.E. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* **2021**, *372*. [CrossRef]
67. Ha, S.; Marchetto, D.J.; Dharur, S.; Asensio, O.I. Topic classification of electric vehicle consumer experiences with transformer-based deep learning. *Patterns* **2021**, *2*, 100195. [CrossRef] [PubMed]
68. Raja, A.; Prema, S. Unlocking Customer Insights: A Hybrid SVM-GPT Transformer Model for User Engagement. *Int. J. Intell. Syst. Appl. Eng.* **2024**, *12*, 441–448.
69. Liu, Y.; Hou, J.; Zhao, W. Deep Learning and User Consumption Trends Classification and Analysis Based on Shopping Behavior. *J. Organ. End User Comput.* **2024**, *36*, 1–23. [CrossRef]
70. Peng, Z.; Rong, Y.; Zhu, T. Transformer-based choice model: A tool for assortment optimization evaluation. *Nav. Res. Logist.* **2024**, *71*, 854–877. [CrossRef]
71. Zhang, Y.; Du, J.; Ma, X.; Wen, H.; Fortino, G. Aspect-Based Sentiment Analysis for User Reviews. *Cogn. Comput.* **2021**, *13*, 1114–1127. [CrossRef]
72. Wang, Y.; Willis, E.; Yeruva, V.K.; Ho, D.; Lee, Y. A case study of using natural language processing to extract consumer insights from tweets in American cities for public health crises. *BMC Public Health* **2023**, *23*, 935. [CrossRef]
73. Zhang, H.; Mu, X.; Li, G.; Xu, Z.; Yu, X.; Ma, J. Mannequin2Real: A Two-Stage Generation Framework for Transforming Mannequin Images Into Photorealistic Model Images for Clothing Display. *IEEE Trans. Consum. Electron.* **2024**, *70*, 2773–2783. [CrossRef]

74. Sohn, K.; Sung, C.E.; Koo, G.; Kwon, O. Artificial intelligence in the fashion industry: Consumer responses to generative adversarial network (GAN) technology. *Int. J. Retail Distrib. Manag.* **2021**, *49*, 61–80. [CrossRef]
75. Mamta, K.; Sangwan, S. AaPiDL: An ensemble deep learning-based predictive framework for analyzing customer behaviour and enhancing sales in e-commerce systems. *Int. J. Inf. Technol.* **2024**, *16*, 3019–3025. [CrossRef]
76. Wei, S.; Zhou, X.; An, X.; Yang, X.; Xiao, Y. A heterogeneous E-commerce user alignment model based on data enhancement and data representation. *Expert Syst. Appl.* **2023**, *228*, 120258. [CrossRef]
77. Hasumoto, K.; Goto, M. Predicting customer churn for platform businesses: Using latent variables of variational autoencoder as consumers' purchasing behavior. *Neural Comput. Appl.* **2022**, *34*, 18525–18541. [CrossRef]
78. Mena, G.; Coussement, K.; De Bock, K.W.; De Caigny, A.; Lessmann, S. Exploiting time-varying RFM measures for customer churn prediction with deep neural networks. *Ann. Oper. Res.* **2024**, *339*, 765–787. [CrossRef]
79. Crisp, D.; Newsted, J.; Brendon, B.; Barnes, D.; Hayes, C.; Prantner, J. Customising generative AI: Harnessing document retrieval and fine-tuning alternatives for dynamic marketing insights. *Appl. Mark. Anal.* **2024**, *10*, 18–31. [CrossRef]
80. Hermann, E.; Puntoni, S. Artificial intelligence and consumer behavior: From predictive to generative AI. *J. Bus. Res.* **2024**, *180*, 114720. [CrossRef]
81. Wu, X.; Li, P.; Zhao, M.; Liu, Y.; Crespo, R.G.; Herrera-Viedma, E. Customer churn prediction for web browsers. *Expert Syst. Appl.* **2022**, *209*, 118177. [CrossRef]
82. Ahmad, N.; Awan, M.J.; Nobanee, H.; Zain, A.M.; Naseem, A.; Mahmoud, A. Customer Personality Analysis for Churn Prediction Using Hybrid Ensemble Models and Class Balancing Techniques. *IEEE Access* **2024**, *12*, 1865–1879. [CrossRef]
83. Kim, H.; Moon, N. TN-GAN-Based Pet Behavior Prediction through Multiple-Dimension Time-Series Augmentation. *Sensors* **2023**, *23*, 4157. [CrossRef]
84. Azadravesh, H.; Sheibani, R.; Forghani, Y. Predicted consumer buying behavior in neural marketing based on convolutional neural network and short-term long-term memory. *Multimed. Tools Appl.* **2024**, 1–7. [CrossRef]
85. Gandhudi, M.; Alphonse, P.J.A.; Velayudham, V.; Nagineni, L.; Gangadharan, G.R. Explainable causal variational autoencoders based equivariant graph neural networks for analyzing the consumer purchase behavior in E-commerce. *Eng. Appl. Artif. Intell.* **2024**, *136*, 108988. [CrossRef]
86. Gao, Y.; Ma, Q.; Cao, J.; Wang, Y.; Yang, X.; Xu, Q.; Liang, Q.; Sun, Y. Recent advances in microfluidic-aided chitosan-based multifunctional materials for biomedical applications. *Int. J. Pharm.* **2021**, *600*, 120465. [CrossRef]
87. Elsayed, M.; Zafar, K.; Esa, Y.; Soujad, T.; Mohamed, A. Impact of 100% vehicle electrification on the distribution grid in dense urban regions. *Energy Rep.* **2024**, *11*, 5315–5322. [CrossRef]
88. Theocharis, A.; Hamanee, S. Battery Storage at the Secondary Distribution Electricity Grid by Investigating End-Users Load Demand Measurements. *Energies* **2022**, *15*, 2743. [CrossRef]
89. Machado, L.F.M.; Santo, S.G.D.; Junior, G.M.; Itiki, R.; Manjrekar, M.D. Multi-Source Distributed Energy Resources Management System Based on Pattern Search Optimal Solution Using Nonlinearized Power Flow Constraints. *IEEE Access* **2021**, *9*, 30374–30385. [CrossRef]
90. Ismail, A.G.; El-Dabah, M.A.; Nassar, I.A. Enhancement of Electrical Distribution Networks Performance Using the Load Management Methodology. *Energy Rep.* **2020**, *6*, 2066–2074. [CrossRef]
91. Al-Bahrani, L.T.; Seyedmahmoudian, M.; Horan, B.; Stojcevski, A. Distribution transformer load behavior, burden, and characteristics of residential consumers: A case study of Baghdad City. *Energy Build.* **2020**, *210*, 109693. [CrossRef]
92. McGarry, C.; Dixon, J.; Elders, I.; Galloway, S. A high-resolution geospatial and socio-technical methodology for assessing the impact of electrified heat and transport on distribution network infrastructure. *Sustain. Energy Grids Netw.* **2023**, *35*, 101118. [CrossRef]
93. Wang, Z.; Hong, T. Generating realistic building electrical load profiles through the Generative Adversarial Network (GAN). *Energy Build.* **2020**, *224*, 110299. [CrossRef]
94. Xu, W.; Jiang, C.; Zhang, Q.; Zheng, J. An intelligent non-intrusive load monitoring model based on power encoding and convolutional state modules. *Meas. Sci. Technol.* **2024**, *35*, 086210. [CrossRef]
95. Tao, Y.; Qiu, J.; Lai, S.; Wang, Y.; Sun, X. Reserve Evaluation and Energy Management of Micro-grids in Joint Electricity Markets Based on Non-Intrusive Load Monitoring. *IEEE Trans. Ind. Appl.* **2023**, *59*, 207–219. [CrossRef]
96. Langevin, A.; Cheriet, M.; Gagnon, G. Efficient deep generative model for short-term household load forecasting using non-intrusive load monitoring. *Sustain. Energy Grids Netw.* **2023**, *34*, 101006. [CrossRef]
97. Amer, A.; Shaban, K.; Gaouda, A.; Massoud, A. Home energy management system embedded with a multi-objective demand response optimization model to benefit customers and operators. *Energies* **2021**, *14*, 257. [CrossRef]
98. Sarker, I.H. Machine Learning: Algorithms, Real-World Applications and Research Directions. *SN Comput. Sci.* **2021**, *2*, 160. [CrossRef] [PubMed]
99. Whang, S.E.; Roh, Y.; Song, H.; Lee, J.-G. Data collection and quality challenges in deep learning: A data-centric AI perspective. *VLDB J.* **2023**, *32*, 791–813. [CrossRef]
100. Zhang, E.Y.; Cheok, A.D.; Pan, Z.; Cai, J.; Yan, Y. From Turing to Transformers: A Comprehensive Review and Tutorial on the Evolution and Applications of Generative Transformer Models. *Sci* **2023**, *5*, 46. [CrossRef]

101. Liang, L.; Wenjun, W.; Dichen, D.; Fei, Z.; Yizhou, Y.; Yaokuan, M. A resource scheduling algorithm of cloud computing based on energy efficient optimization methods. In Proceedings of the 2012 International Green Computing Conference (IGCC), San Jose, CA, USA, 4–8 June 2012; pp. 1–6.
102. Madaan, G.; Asthana, S.K.; Kaur, J. Generative AI: Applications, Models, Challenges, Opportunities, and Future Directions. In *Generative AI and Implications for Ethics, Security, and Data Management*; Gomathi Sankar, J., David, A., Eds.; IGI Global: Hershey, PA, USA, 2024; pp. 88–121. [CrossRef]
103. Díaz-Rodríguez, N.; Del Ser, J.; Coeckelbergh, M.; López de Prado, M.; Herrera-Viedma, E.; Herrera, F. Connecting the dots in trustworthy Artificial Intelligence: From AI principles, ethics, and key requirements to responsible AI systems and regulation. *Inf. Fusion* **2023**, *99*, 101896. [CrossRef]
104. van Esch, P.; Cui, Y.; Heilgenberg, K. Using Artificial Intelligence (AI) to Implement Diversity, Equity and Inclusion (DEI) into Marketing Materials: The 'CONSIDER' Framework. *Australas. Mark. J.* **2024**, *32*, 250–262. [CrossRef]
105. Barredo Arrieta, A.; Díaz-Rodríguez, N.; Del Ser, J.; Bennetot, A.; Tabik, S.; Barbado, A.; Garcia, S.; Gil-Lopez, S.; Molina, D.; Benjamins, R.; et al. Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Inf. Fusion* **2020**, *58*, 82–115. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.